

User Stories and Acceptance Criteria

AI-Powered Customer Support Ticket Triage - SaaS Delivery Simulation

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Document Type: Agile User Stories, Acceptance Criteria, and UAT Readiness Pack

Project Context: Independent portfolio simulation for an AI-enabled customer experience SaaS workflow

This document converts the product and business requirements of the customer support ticket triage workflow into a structured story backlog. It is designed to demonstrate delivery thinking, requirement clarity, prioritization, acceptance criteria writing, and handover readiness for a SaaS implementation environment.

1. Executive Overview

This User Stories and Acceptance Criteria document defines how the simulated AI-enabled support triage workflow should behave from the perspective of support agents, managers, implementation teams, product teams, and customer success users. The workflow focuses on reducing manual ticket sorting, improving consistency in prioritization, and ensuring that customer issues are routed to the correct support teams.

The project is intentionally structured like a delivery artefact used in a SaaS implementation environment: each user story links a role, a business need, an expected outcome, acceptance criteria, validation checks, dependencies, and edge-case handling. The scope remains a portfolio simulation and does not claim production deployment, real client implementation, or model training.

2. Product Context and Delivery Objective

Area	Professional Definition
Business problem	UrbanCart, a mock e-commerce client, receives high volumes of customer support tickets that require manual reading, categorization, prioritization, and routing.
Delivery objective	Create a structured SaaS-style workflow specification that demonstrates how an AI-assisted triage capability could support customer support operations.
Primary outcome	Support tickets should receive an issue category, priority level, routing team, and manual review flag where confidence or clarity is insufficient.
Target users	Support Agent, Support Manager, Implementation Team Member, Product Team Member, and Customer Success Team Member.
Document purpose	Translate business and product requirements into testable user stories and acceptance criteria suitable for implementation planning, demo preparation, and UAT discussion.

3. Story Writing Standard

The user stories follow a standard Agile requirement pattern: As a [persona], I want [capability], so that [business value]. Acceptance criteria are written in a Given-When-Then format so that product, implementation, and technology teams can validate expected behavior without ambiguity.

Element	Purpose
Persona	Identifies who receives value from the feature or process step.
Need	Defines the functional or operational capability required by the user.
Business value	Explains why the capability matters to the customer support workflow.
Acceptance criteria	Defines observable conditions that must be true for the story to be considered complete.
Exception handling	Documents expected behavior for unclear, incomplete, or unsupported ticket scenarios.

4. Personas and Operating Responsibilities

Persona	Responsibilities	Needs from Workflow
Support Agent	Handles customer tickets, confirms classification output, responds to customers, and updates ticket status.	Clear category, priority, routing team, ticket context, and manual review option.
Support Manager	Monitors high-priority tickets, backlog volume, routing consistency, escalations, and service performance.	Visibility into urgent tickets, category distribution, exceptions, and operational risks.
Implementation Team Member	Converts requirements into workflow configuration, validation steps, and delivery artefacts.	Traceable requirements, routing rules, acceptance criteria, and open question tracking.
Product Team Member	Reviews feature scope, priority model, user roles, and product behavior expectations.	Clear product requirements, story priority, dependencies, and quality gates.
Customer Success Team Member	Supports client demos, onboarding, training, feedback collection, and adoption.	Demo-ready workflow, training guide, simple user language, and adoption support material.

5. Epic-Level Backlog

Epic ID	Epic	Description	Outcome
EP-01	Ticket Classification	Automatically map incoming ticket content to a predefined issue category.	Support agents spend less time manually sorting tickets.
EP-02	Priority Assignment	Assign High, Medium, or Low priority based on urgency, risk, and customer impact.	Critical tickets are easier to identify and resolve faster.
EP-03	Team Routing	Route tickets to the correct internal support team based on category and issue type.	Tickets reach the right team with fewer handoff delays.
EP-04	Manual Review and Exception Handling	Flag unclear tickets for human review when classification is uncertain or unsupported.	Incorrect automated routing is reduced.
EP-05	Demo, Training, and Adoption	Prepare a simple explanation of how users should understand and operate the workflow.	Client-facing users can understand the solution without deep technical knowledge.

6. Prioritization Model

Stories are prioritized using a simplified MoSCoW model to show delivery sequencing and business importance. Must-have stories represent the minimum viable workflow required for a credible ticket triage simulation. Should-have stories strengthen operational reliability. Could-have stories improve reporting and adoption but are not required for the initial demonstration.

Priority	Definition	Stories
Must Have	Required for the core workflow to make sense and be demonstrable.	Ticket Classification, Priority Assignment, Team Routing
Should Have	Important for quality, reliability, and implementation confidence.	Manual Review, Exception Handling, Acceptance Validation
Could Have	Useful for client adoption and presentation polish.	Demo and Training Support, Manager Filters, Feedback Capture
Won't Have	Explicitly outside this simulation scope.	Real model training, live SaaS deployment, real customer data processing

7. Detailed User Stories

User Story 1: Ticket Classification

Persona: Support Agent | **MoSCoW Priority:** Must Have

As a support agent, I want incoming tickets to be automatically categorized so that I can focus on resolving customer issues instead of manually sorting tickets.

Business value	Reduces repetitive manual sorting and improves category consistency across refund, delivery, payment, damaged product, cancellation, login, and general query tickets.
Primary dependency	Predefined ticket taxonomy, sample ticket data, and category mapping rules must be available.
Validation metrics	Percentage of categorized tickets, manual review rate, classification consistency, and agent review effort.
Exception handling	Tickets with unclear wording, multiple unrelated issues, or unsupported topics should be flagged for manual review.

Acceptance Criteria - BDD Format

Scenario	Given	When	Then	Priority
Known category	A ticket contains language related to refund, delivery delay, payment issue, damaged product, cancellation, login issue, or general query.	The ticket is processed by the workflow.	The ticket receives exactly one primary category from the predefined list.	High
Agent visibility	A ticket has been categorized.	A support agent opens the ticket view.	The assigned category is visible along with the original customer message.	High
Unsupported category	A ticket does not match any predefined category.	The workflow attempts classification.	The ticket is marked for Manual Review instead of being forced into an inaccurate category.	Medium

User Story 2: Priority Assignment

Persona: Support Manager | MoSCoW Priority: Must Have

As a support manager, I want urgent tickets to be flagged as high priority so that critical customer issues can be handled faster.

Business value	Improves operational visibility by highlighting high-impact cases such as payment failure, repeated complaint, damaged product, angry tone, and escalation requests.
Primary dependency	Priority rules must be aligned with issue type, urgency indicators, and customer-impact keywords.
Validation metrics	High-priority detection rate, average response focus, number of escalations surfaced, and manager review readiness.
Exception handling	Where priority indicators conflict, the workflow should select the higher priority and add a review note.

Acceptance Criteria - BDD Format

Scenario	Given	When	Then	Priority
High-priority issue	A ticket includes payment failure, money deducted, repeated complaint, damaged product, angry tone, or escalation wording.	The workflow evaluates priority.	The ticket is assigned High Priority and routed for faster attention.	High
Medium-priority issue	A ticket relates to refund pending, delayed delivery, or cancellation request without escalation wording.	The workflow evaluates priority.	The ticket is assigned Medium Priority.	High
Low-priority issue	A ticket is a general question or basic account/status request.	The workflow evaluates priority.	The ticket is assigned Low Priority.	Medium
Manager filtering	Tickets have assigned priorities.	A manager reviews the ticket queue.	High Priority tickets can be identified separately for operational follow-up.	Medium

User Story 3: Team Routing

Persona: Implementation Team Member | **MoSCoW Priority:** Must Have

As an implementation team member, I want each ticket category to map to a support team so that tickets are routed consistently.

Business value	Reduces handoff delays and creates a clear operating model for support teams handling billing, logistics, payments, returns, order management, technical support, and general support.
Primary dependency	Category-to-team mapping must be defined and reviewed before the demo workflow is considered complete.
Validation metrics	Correct routing coverage, handoff clarity, support team mapping completeness, and routing exception rate.
Exception handling	If a ticket contains more than one issue, primary routing should follow the highest-impact issue and secondary context should be preserved.

Acceptance Criteria - BDD Format

Scenario	Given	When	Then	Priority
Refund routing	A ticket is categorized as Refund.	Routing is applied.	The ticket routes to Billing & Refund Team.	High
Delivery routing	A ticket is categorized as Delivery Delay.	Routing is applied.	The ticket routes to Logistics Team.	High
Payment routing	A ticket is categorized as Payment Issue.	Routing is applied.	The ticket routes to Payments Team.	High
Login routing	A ticket is categorized as Login Issue.	Routing is applied.	The ticket routes to Technical Support Team.	Medium

User Story 4: Manual Review and Exception Handling

Persona: Support Agent | **MoSCoW Priority:** Should Have

As a support agent, I want unclear tickets to be marked for manual review so that incorrect routing can be avoided.

Business value	Prevents the workflow from over-automating ambiguous cases and keeps a human review path for unclear, mixed, incomplete, or unsupported tickets.
Primary dependency	Manual Review status must exist as an exception state with a clear default team owner.
Validation metrics	Manual review volume, unclear ticket handling rate, correction frequency, and exception documentation quality.
Exception handling	Manual Review should be treated as a controlled fallback, not a failure of the workflow.

Acceptance Criteria - BDD Format

Scenario	Given	When	Then	Priority
Unclear content	A ticket lacks enough information to identify category or priority confidently.	The workflow processes the ticket.	The ticket is marked Manual Review and routed to General Support Team.	High
Agent correction	A support agent reviews a Manual Review ticket.	The agent identifies the correct category.	The category can be updated in the documentation workflow.	Medium
Incomplete message	A customer submits a vague message such as "help me".	The workflow evaluates the ticket.	The ticket is not forced into a specific category and is marked for human review.	High

User Story 5: Demo and Training Readiness

Persona: Customer Success Team Member | **MoSCoW Priority:** Could Have

As a customer success team member, I want a simple demo flow and training guide so that client users can understand how to use the workflow.

Business value	Improves adoption by explaining the workflow in client-friendly language and preparing clear demonstration material for non-technical users.
Primary dependency	Demo script, training guide, sample tickets, and output explanation must be available.
Validation metrics	Demo clarity, stakeholder understanding, training completeness, and ability to explain business value.
Exception handling	If a client asks about real AI model accuracy, clarify that this portfolio project simulates workflow behavior rather than claiming production model performance.

Acceptance Criteria - BDD Format

Scenario	Given	When	Then	Priority
Demo walkthrough	A sample support ticket is selected.	The demo is presented.	The presenter can explain ticket intake, category, priority, routing, and support usage clearly.	Medium
Training material	A new support user reviews the guide.	They read the categories and priority definitions.	They understand how to interpret workflow output without technical background.	Medium
Client questions	A stakeholder asks why manual review exists.	The presenter explains exception handling.	The stakeholder understands that manual review reduces incorrect automation.	Medium

8. Requirement Traceability Matrix

Story ID	Mapped Requirement	Primary Persona	Business Outcome	Validation Artefact
US-01	Classify tickets into predefined categories.	Support Agent	Reduced manual sorting effort.	Category mapping and sample ticket validation.
US-02	Assign priority based on urgency and issue type.	Support Manager	Faster identification of critical customer issues.	Priority rules and high-priority sample cases.
US-03	Route tickets to appropriate support teams.	Implementation Team Member	Consistent routing and fewer handoff delays.	Routing matrix and team mapping table.
US-04	Support manual review for unclear tickets.	Support Agent	Reduced incorrect classification and safer automation.	Manual review scenarios and exception cases.
US-05	Support demo and training readiness.	Customer Success Team Member	Improved client adoption and solution understanding.	Demo script, training guide, and sample walkthrough.

9. Release Story Map

Release Layer	Included Stories	Purpose
MVP Demo	US-01 Ticket Classification, US-02 Priority Assignment, US-03 Team Routing	Demonstrates the minimum viable SaaS triage workflow.
Reliability Layer	US-04 Manual Review and Exception Handling	Adds safer handling of unclear tickets and improves operational credibility.
Adoption Layer	US-05 Demo and Training Readiness	Prepares the project for stakeholder explanation, client-facing walkthrough, and interview discussion.

10. Definition of Ready and Definition of Done

Category	Definition
Definition of Ready	A story is ready when the persona, business value, dependencies, acceptance criteria, sample data requirement, and expected output are clearly documented.
Definition of Done	A story is complete when acceptance criteria are testable, expected behavior is documented, exception handling is defined, and the story is traceable to the workflow objective.
Documentation Quality Gate	The story should be understandable by product, implementation, and support stakeholders without requiring additional verbal explanation.
Demo Quality Gate	At least one sample ticket should be available to demonstrate the story outcome during a walkthrough.

11. UAT Scenario Pack

Scenario ID	Input Example	Expected Category	Expected Priority	Expected Routing
UAT-01	My payment was deducted but the order was not placed. Please help urgently.	Payment Issue	High	Payments Team
UAT-02	I have not received my order even after the expected delivery date.	Delivery Delay	Medium	Logistics Team
UAT-03	The product I received is damaged and I need a replacement immediately.	Damaged Product	High	Returns & Replacement Team
UAT-04	I cannot login to my account even after resetting my password.	Login Issue	Low	Technical Support Team
UAT-05	Help me with my order.	Manual Review	Medium or Low based on review	General Support Team

12. Edge Cases and Clarification Rules

Edge Case	Expected Handling	Reason
Multiple issues in one ticket	Assign the primary category based on highest customer impact and preserve secondary context in notes.	Prevents loss of important context while keeping routing clear.
Angry tone without clear issue	Flag for Manual Review and mark priority as Medium or High depending on escalation language.	Avoids inaccurate categorization while still surfacing urgency.
Payment and delivery issue together	Prioritize Payment Issue if money is deducted or transaction failed; otherwise route based on delivery problem.	Payment issues usually create higher customer-impact risk.
Very short message	Flag for Manual Review.	Insufficient data can lead to wrong routing.
Unsupported language or unclear text	Route to General Support Team with Manual Review flag.	Maintains safe fallback for unsupported inputs.

13. Non-Functional Acceptance Criteria

Area	Acceptance Standard
Explainability	The workflow should provide category, priority, and routing output in a way that support users can understand.
Usability	Support users should not need technical background to interpret ticket category and priority.
Data Safety	The simulation should avoid real customer data and use mock sample tickets only.
Maintainability	Category and routing rules should be documented clearly enough for future updates.
Traceability	Each core behavior should map back to a documented user story and acceptance criterion.

14. Assumptions, Constraints, and Out-of-Scope Items

Type	Details
Assumptions	Tickets are text-based, categories are predefined, and the simulation uses mock data.
Constraints	The project is documentation-led and does not represent a production SaaS deployment.
Out of Scope	Real AI model training, live CRM integration, real customer data processing, production security controls, and SLA-backed enterprise rollout.
Professional Positioning	The artefact demonstrates AI delivery understanding, SaaS workflow planning, requirement clarity, and client-facing documentation readiness.

15. Handover and Interview Positioning

This document can be used as a portfolio artefact to explain how business needs are converted into user stories, acceptance criteria, validation scenarios, and demo-ready requirements. It supports an entry-level AI Delivery or SaaS Implementation discussion by showing that the candidate understands client requirements, cross-functional coordination, product documentation, and structured delivery outputs.

Suggested interview explanation: I created user stories and acceptance criteria for an AI-powered customer support ticket triage simulation. The goal was to understand how SaaS delivery teams translate a client problem into documented requirements, workflow behavior, routing rules, priority logic, UAT scenarios, and demo/training support material. This helped me connect technical project work with client-facing implementation and product delivery practices.